Mini Ecommerce Application using Microservices Architecture

Technologies Used:
Angular | Spring Boot | MySQL | Kafka | Eureka | Config Server | API Gateway

Submitted By:
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Organization / Course:
Capstone Project — Microservices Based Ecommerce Application

Problem Statement:

.Traditional monolithic ecommerce systems face challenges such as:
Poor scalability
.Difficult maintenance
.Tight coupling between modules
.Single point of failure
.To overcome these issues, a Microservices-based ecommerce application is designed.

Objectives

Develop Microservices-based ecommerce system
Implement CQRS architecture
Implement SAGA using Kafka
Implement Circuit Breaker for fault tolerance
Implement centralized configuration using Config Server

Scope Included

User Management
Product Management
Order Management
Cart functionality
Microservices architecture

Excluded

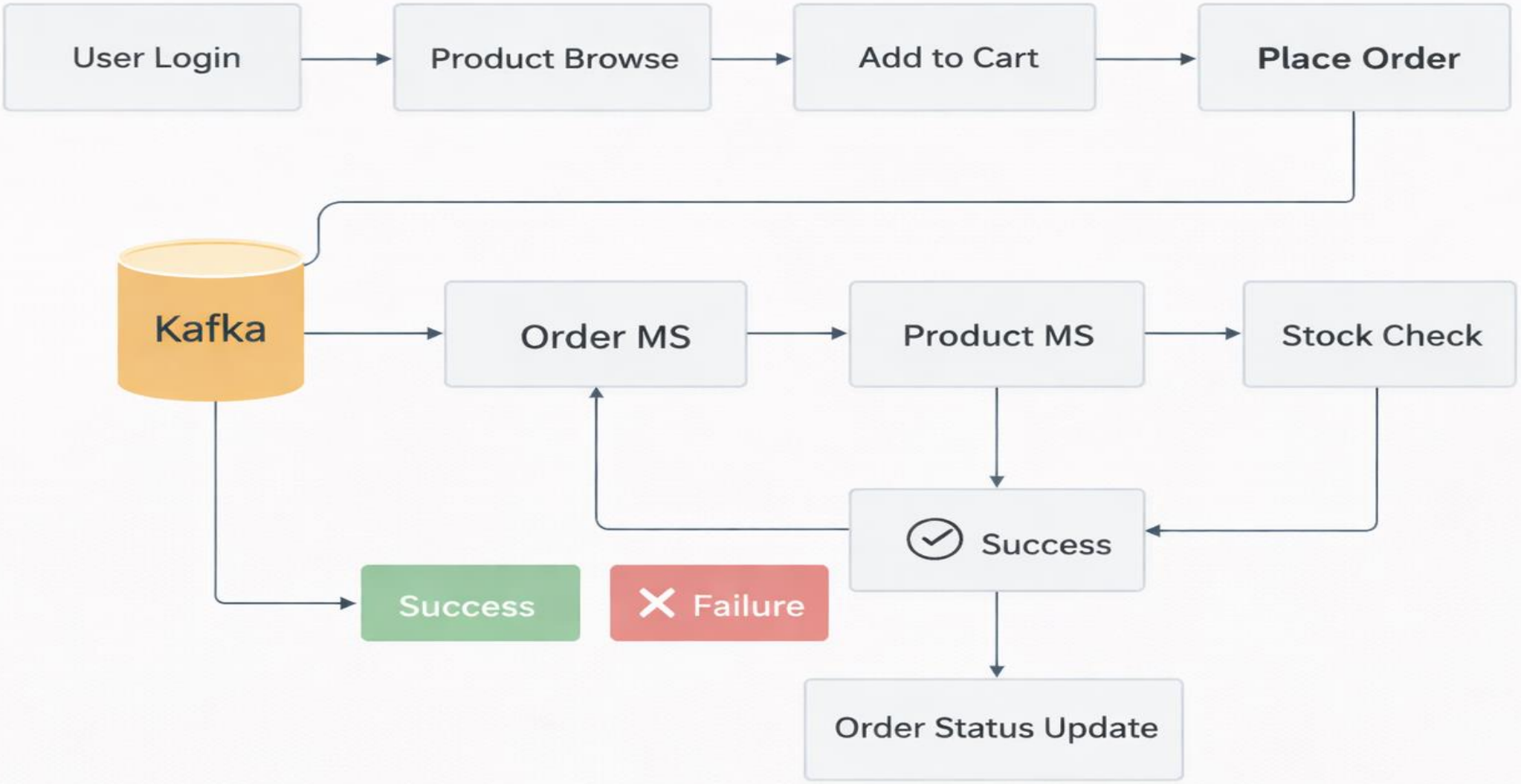
Payment Gateway integration
External vendor integration
Email notification system

Proposed Solution / System Overview

System Architecture:
Microservices:
User MS (CQRS → Read + Write)
Product MS
Order MS
API Gateway
Service Registry (Eureka)
Config Server

Key Features:
Role Based Access (Admin / Customer)
Product Catalog
Shopping Cart
Order Processing
Distributed Transaction (SAGA)
Circuit Breaker

Workflow:
User Login → Product Browse → Add to Cart → Place Order
→ Order MS → Kafka → Product MS → Stock Check
→ Success / Failure → Order Status Update


User Login
Product Browse
Add to Cart
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Order MS
Product MS
Stock Check
Success
Success
Failure
Order Status Update

Technology Stack & Implementation

Frontend

Angular
Bootstrap
HTML / CSS / TypeScript

Backend

Spring Boot
Spring Cloud
Spring Data JPA

Database

MySQL

Tools

- Eureka Server
- Config Server
- API Gateway
- Kafka
- Circuit Breaker (Resilience4j)
- Swagger
- Zipkin

Login to

Shop From Us

Email address

Enter email

Password

Password

Remember me

Forgot Password?

Login

or sign in with

G f I

Don't have an account? Register

+ Add Product

Shivang Shivhare ▼

All Categories

Search product...

Search

Price

Quantity

Product	Category	Price	UOM	Stock	Status	Description	Edit	Delete
Fire-Boltt Rise Smart Watch	Watch	₹ 1399	Piece	24	In Stock	Fire-Boltt Rise Smart Watch	Edit	Delete
Ikigai	Books	₹ 320	Piece	96	In Stock	Ikigai: The Japanese secret to a l...	Edit	Delete
NutriPro Juicer Mixer Grinder	Electronics	₹ 1599	Piece	57	In Stock	NutriPro Juicer Mixer Grinder fo...	Edit	Delete
XIAOMI 17 Ultra	Mobile Phones	₹ 139999	Piece	7	In Stock	XIAOMI 17 Ultra best in segmen...	Edit	Delete
HP 15, AMD Ryzen 5 7520U	Laptop	₹ 41346	Piece	4	In Stock	HP 15, AMD Ryzen 5 7520U	Edit	Delete

Prev Next

Shop With Us

All

Search...

Search

Orders

Cart (0)

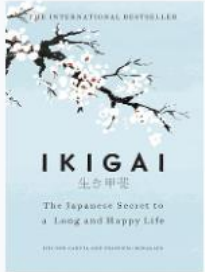
User

Fire-Boltt Rise Smart Watch

Watch

₹ 1399

Add to Cart



Ikigai

Books

₹ 320

Add to Cart

NutriPro Juicer Mixer Grinder

Electronics

₹ 1599

Add to Cart

XIAOMI 17 Ultra

Mobile Phones

₹ 139999

Add to Cart

HP 15, AMD Ryzen 5 7520U

Laptop

₹ 41346

Add to Cart

DOT & KEY Bright Sunscreen

Skincare

₹ 387

Add to Cart

Plum Body Lotion

Skincare

₹ 243

Out of Stock

Fire-Boltt Rise Smart Watch

Watch

₹ 1299

Add to Cart

← Back

Shopping Cart

Welcome, sachin kumar

XIAOMI 17 Ultra

₹139999

In Stock

- 1 + Remove

NutriPro Juicer Mixer Grinder

₹1599

In Stock

- 1 + Remove

Total Items: 2

Total Amount: ₹141598

Proceed to Buy

Key Results:

. Scalable architecture
. Fault tolerant system
. Loose coupling between services
. Distributed transaction handling

Flow User

Login → Add Product → Add to Cart → Place Order Order Created → Product Stock Reduced → Order Completed

Conclusion & Future Scope

Conclusion

- Microservices based ecommerce system successfully developed
- CQRS architecture implemented
- Kafka based SAGA implemented
- Circuit breaker implemented
- Distributed tracing implemented

Challenges Faced

- Kafka configuration
- Microservice communication
- Circuit breaker implementation
- Distributed transaction handling

Limitations

- No payment gateway
- Basic UI design
- No email notification

Future Enhancements

- Payment Gateway Integration
- Email Notification
- JWT Security
- Kubernetes Deployment
- Docker Containerization

Thank you